Chirag Batra

B.Tech CS (Data Science) · Class of 2027

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SUMMARY

Final-year Computer Science (Data Science) student, class of 2027, targeting ML engineering and data science roles. I build ML systems end to end and measure them honestly: real-time fraud detection on AWS, causal uplift modeling and a multi-agent LLM deployment platform. Two data internships, including IBM.

EDUCATION

UPES Dehradun, Bachelor of Technology in Computer Science (Data Science) 2023 – 2027

EXPERIENCE

Project Intern, Data Science, IBM · *Remote* Jun 2026 – Jul 2026

- Built EventLens AI, a full-stack financial risk analytics platform in Python, computing Value-at-Risk and Monte Carlo projections for portfolio exposure.
- Trained a Random Forest model using a Scikit-learn pipeline on 250+ event-days of GDELT news data to predict abnormal returns from news sentiment and volatility features.
- Developed ETL pipelines aggregating news event data with live market pricing into a SQLAlchemy database, delivered through FastAPI REST services and a React reporting dashboard.

Research Intern, Data Analysis, ISVR Aerospace Projects · *Dehradun, India* Jun 2025 – Jul 2025

- Developed Python preprocessing pipelines to clean drift and noise from four sensor telemetry streams, including GPS, IMU, barometric pressure, and temperature data.
- Implemented automated quality control checks that detected altitude-control anomalies in time-series sensor data.
- Conducted comparative analysis across five operational scenarios, quantifying performance trade-offs for technical stakeholders.

PROJECTS

RADAR: Real-time Agentic Detection and Reporting GitHub ↗

Streaming fraud detection on AWS: XGBoost scoring at 0.4 ms median (717 req/s in load tests) with 89.6% recall at 1% FPR on 606k synthetic transactions, beating rules (66.8%) and logistic regression (73.7%). Hourly PSI drift monitoring triggers champion/challenger retraining; a Bedrock tool-use agent investigates flagged transactions and returns schema-validated verdicts. ~60 AWS resources in Terraform, CI with OIDC deploys, 42 tests. Python · XGBoost · AWS (Kinesis, Lambda, Fargate, SQS) · Amazon Bedrock · Terraform · FastAPI

Customer Churn Prediction and Retention Uplift Modeling GitHub ↗

Churn model on a 7,032-customer telecom dataset reaching 0.82 AUC-ROC with XGBoost. Uplift modeling identified persuadable customers, cutting simulated retention campaign spend by 56% while maintaining retention rates. Streamlit and FastAPI dashboard with SHAP explainability; experiments tracked in MLflow. Python · XGBoost · CausalML · SHAP · MLflow · FastAPI · Streamlit

DeploySmith OS GitHub ↗

A portfolio that doubles as a multi-agent deployment platform: LangGraph agents statically profile a GitHub repo, plan its hosting, build it in hardened Docker containers and health-check the result, streaming every step over SSE. Immutable deploys with build logs and one-click rollback; 200+ pytest tests. Python · LangGraph · FastAPI · Docker · React · TypeScript

Proactive Academic Intervention System GitHub ↗

End-to-end ML web app analyzing 5,000+ records to predict at-risk individuals, reaching 0.887 ROC-AUC with class-balanced Logistic Regression. Automated feature engineering from MySQL, a max-heap priority queue for ranking, and sliding-window anomaly detection. Flask backend deployed on Render. Python · Scikit-learn · Flask · MySQL · Render

SKILLS

Languages: Python, SQL, TypeScript, Java, Bash

Machine Learning: Scikit-learn, XGBoost, LightGBM, Feature Engineering, Model Evaluation, Imbalanced Learning (SMOTE), SHAP Explainability, Hyperparameter Tuning, Time-series Analysis

Deep Learning & LLMs: PyTorch, Hugging Face Transformers, LoRA / QLoRA (PEFT), RAG, Embeddings & Vector Search, LangGraph Agents, Tool Use & Structured Output, LLM Evaluation, Claude API, Amazon Bedrock, vLLM

Statistics & Experimentation: Hypothesis Testing, A/B Testing, Causal Inference, Uplift Modeling (CausalML), Drift Detection (PSI), Probability & Statistics

MLOps & Cloud: Docker, AWS (Lambda, Fargate, Kinesis, SQS, DynamoDB, S3), Terraform, GitHub Actions CI/CD, MLflow, Model Serving (FastAPI), Monitoring & Retraining

Data Engineering: Pandas, NumPy, Polars, ETL Pipelines, PostgreSQL / pgvector, MySQL, SQLAlchemy, Data Modeling

Visualization & BI: Matplotlib, Seaborn, Plotly, Streamlit, Tableau, Power BI

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Certifications: Data Science Tools, IBM SkillsBuild; Data Analytics Virtual Experience Program, Deloitte Australia (Forage); Data Structures and Algorithms using Python, Tutedude